



**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)**

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. I - Semester							
ELECTRONICS & COMMUNICATION ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23BS2105	Probability theory and stochastic process	BS	3	0	0	3	30	70	100
B23HS2101	Universal Human Values -II : Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100
B23EC2101	Signals and Systems	PC	3	0	0	3	30	70	100
B23EC2102	Electronic Devices and Circuits	PC	3	0	0	3	30	70	100
B23EC2103	Switching Theory& Logic Design	PC	3	0	0	3	30	70	100
B23EC2104	Electronic Devices and Circuits Lab	PC	0	0	3	1.5	30	70	100
B23EC2105	Switching Theory& Logic Design Lab	PC	0	0	3	1.5	30	70	100
B23EC2106	Data Structures using Python	SEC	0	1	2	2	30	70	100
B23MC2102	Environmental Science	MC	2	0	0	-	30	-	30
TOTAL			16	2	8	20	270	560	830

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2105	BS	3	--	--	3	30	70	3 Hrs.
PROPABILITY THEORY & STOCHASTIC PROCESSES								
(For ECE)								
Course Objectives:								
1.	To introduce the fundamental concepts and theorems of probability theory.							
2.	To introduce fundamental concepts of random variables and their statistical descriptions							
3.	To be aware of the difference between time averages and statistical averages, Stationary and Ergodicity.							
4.	To discuss various types of random processes, their properties, spectral representation and applications.							
5.	To apply elements of stochastic processes for solving problems in real life and identify the concept of Noise as applicable to linear Systems							
Course Outcomes: At the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1.	Describe the axiomatic formulation of modern probability theory.							K2
2.	Recognise Probability Models and functions of Random variables.							K2
3.	Apply the moments and characteristic functions and acquire the concept of Inequalities and probabilistic limits.							K3
4.	Interpret the concept of Random process and determine covariance and spectral density of stationary random processes.							K3
5.	Analyze the response of linear time invariant systems to random inputs.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Probability & Random Variable: Probability introduced through Sets and Relative Frequency: Experiments and Sample Spaces, Discrete and Continuous Sample Spaces, Events, Probability Definitions and Axioms, Joint Probability, Conditional Probability, Total Probability, Bayes Theorem, Independent Events, Random Variable-Definition, Conditions for a Function to be a Random Variable, Discrete, Continuous and Mixed Random Variable, Distribution and Density functions, Properties, Binomial, Poisson, Uniform, Gaussian, Exponential, Rayleigh, Methods of defining Conditioning Event, Conditional Distribution, Conditional Density and their Properties.							
UNIT-II (10 Hrs)	Operations on Single Random Variables – Expectations: Expected Value of a Random Variable, Function of a Random Variable, Moments about							

	the Origin, Central Moments, Variance and Skew, Chebyshev's Inequality, Characteristic Function, Moment Generating Function, Transformations of a Random Variable: Monotonic and Non-monotonic Transformations of Continuous Random Variable.
UNIT-III (10 Hrs)	Operations on multiple random variables Joint Distribution and density Functions and its Properties, Marginal Distribution Functions, Conditional Distribution and Density, Point Conditioning, Conditional Distribution and Density, Interval conditioning, Statistical Independence. Sum of Two Random Variables, Sum of Several Random Variables, Central Limit Theorem, (Proof not expected). Unequal Distribution, Equal Distributions. Expected Value of a Function of Random Variables: Joint Moments about the Origin, Joint Central Moments, Joint Characteristic Functions, Jointly Gaussian Random Variables: Two Random Variables case, N Random Variable case, Properties, Transformations of Multiple Random Variables, Linear Transformations of Gaussian Random Variables
UNIT-IV (10 Hrs)	Random Processes – Temporal Characteristics: The Random Process Concept, Classification of Processes, Deterministic and Nondeterministic Processes, Distribution and Density Functions, concept of Stationarity and Statistical Independence. First-Order Stationary Processes, Second Order and Wide-Sense Stationarity, (N-Order) and Strict-Sense Stationarity, Time Averages and Ergodicity, Mean-Ergodic Processes, Correlation-Ergodic Processes, Autocorrelation Function and Its Properties, Cross-Correlation Function and Its Properties, Covariance Functions, Gaussian Random Processes, Poisson Random Process. Random Processes – Spectral Characteristics: The Power Spectrum: Properties, Relationship between Power Spectrum and Autocorrelation Function, The Cross-Power Density Spectrum, Properties, Relationship between Cross-Power Spectrum and Cross-Correlation Function
UNIT-V (10 Hrs)	Linear Systems and Random Noise Processes: The general linear system, linear time invariant system, time invariant system transfer function, idealized system, causal and stable Systems, Response of Linear Systems to Random signals, Spectral characteristics of system Response, autocorrelation and cross correlation functions of the input and output Response, Low-pass and Bandpass Noise Representation, narrow band process.
Textbooks:	
1.	Probability Theory and Random Signal Principles, PEYTON Z.Peebles, Tata McGraw Hill Publishers.
2.	Probability Theory and Stochastic Processes, Y.Mallikarjuna Reddy, Universities press.
Reference Books:	
1.	Probability Theory and Random Processes, S. P. Eugene Xavier, S. Chand and Co. New Delhi, 1998 (2 nd Edition).

2.	Signal Analysis, Papoulis, McGraw Hill N. Y., 1977.
3.	B.P. Lathi, “Signals, Systems & Communications”, B.S. Publications, 2003.
e-Resources	
1.	https://nptel.ac.in/courses/111/102/111102111/
2.	https://nptel.ac.in/courses/117/105/117105085/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2101	HS	2	1	--	3	30	70	3 Hrs.
UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to all Programmes of Engineering)								
Course Objectives: The objective of this course is to make the student aware of								
1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.							
2	Harmony in the human being, family, society and nature/existence							
3	Holistic perspective towards life, profession and happiness							
Course Outcomes: At the end of this course student will be able to								
S. No.	Outcome							Knowledge Level
1	Explain the role of value education in achieving basic human aspirations.							K2
2	Summarize needs to obtain harmony in self(I).							K2
3	Describe criteria for human-human relationship and harmony in society							K2
4	Explain four orders of nature and our existence							K2
5	Interpret significance of harmony in holistic development							K2
SYLLABUS								
UNIT-I (9 Hrs)	Introduction to Value Education: Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance-understanding and living in harmony at various levels.							
UNIT-II (9 Hrs)	Harmony in the Human Being: Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility).							

	The Body as an Instrument of the Self (I)' (I being the doer, seer and enjoyer). Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of 'I' and harmony in 'I'). Programme to ensure self-regulation(<i>Sanyam</i>) and Health(<i>Swasth</i>)- Exploring Harmony of Self (I) with the Body.
UNIT-III (9 Hrs)	Harmony in the Family and Society: Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV (6 Hrs)	Harmony in the Nature/Existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V (9 Hrs)	Implications of the Holistic Understanding – a Look at Professional Ethics: Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.
Text Books	
1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN

	978-93-87034-53-2
Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews
7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf

Estd. 1980

ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2101	PC	3	--	--	3	30	70	3 Hrs.
SIGNALS AND SYSTEMS								
(For ECE)								
Course Objectives:								
1.	To introduce the fundamental concepts and techniques associated with the understanding of signals and systems.							
2.	To familiarize with techniques suitable for analyzing both continuous-time and discrete-time LTI systems using transforms.							
3.	To familiarize with development of the mathematical skills to solve problems involving convolution, filtering, and sampling.							
Course Outcomes: Student will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate the basic concepts of signals and systems							K2
2.	Analyze the spectral characteristics of periodic and aperiodic signals using Fourier analysis							K4
3.	Analyze the system properties based on impulse response and Fourier analysis for Continuous Time and Discrete Time systems							K4
4.	Apply Laplace transforms for analyzing continuous time 1 st and 2 nd order systems							K3
5.	Apply z- transforms for analyzing discrete time 1 st and 2 nd order systems and Explain the process of sampling & effects of under sampling							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Continuous –Time and Discrete –Time signals and systems: Continuous Time and Discrete Time signals, Energy and Power Signals, Periodic Signals, Even and odd Signals, continuous time complex Exponential and Sinusoidal Signals, Discrete time complex Exponential and Sinusoidal Signals and their Periodicity , The Unit Impulse and Unit step Functions, Continuous –Time and Discrete –Time Systems, Interconnections of Systems, Basic System Properties, Continuous–Time and Discrete Time LTI Systems –Introduction to Convolution, The Graphical interpretation of Convolution Integral & Convolution Sum ,Singularity Functions.							
UNIT-II (10 Hrs)	Fourier Series Representation of Periodic Signals: Introduction, Fourier Series Representation of continuous time Periodic Signals (Complex Exponential and Trigonometric Fourier Series only), convergence of the Fourier Series, Properties of continuous time Fourier Series.							

UNIT-III (10 Hrs)	Continuous and Discrete time Fourier Transform Introduction: Fourier Representation of continuous time aperiodic signals, The continuous time Fourier Transform, Properties of the continuous time Fourier Transform, The Fourier Transform for periodic signals, Systems characterized by linear constant coefficient differential equations. Discrete time Fourier Transform, Properties of the Discrete time Fourier Transform. Related problems. Introduction to Hilbert Transform.
UNIT-IV (10 Hrs)	Laplace Transform Introduction, The Laplace Transform, the region of convergence for Laplace Transforms and its properties, The Inverse Laplace Transform, Properties of Laplace Transforms, Relation between Laplace Transform & Fourier transform, The initial and Final value theorems, Analysis and characterization of LTI systems using the Laplace Transforms.
UNIT-V (10 Hrs)	Sampling Theorem and Z-transform: Introduction to Sampling Theorem, Sampling Theorem for Low pass signals, reconstruction of a signal from its samples using interpolation, discussion on Oversampling, Critical sampling and Under sampling (aliasing). The Z-Transform (Bilateral and unilateral, The Inverse Z-Transform, Properties of Z-Transform, ROC and its properties, Relation between Z -Transform and DTFT, Initial and Final Value theorems, Analysis and characterization of LTI systems using the Z-Transforms.
Textbooks:	
1.	Signals, Systems&Communications-B.P.Lathi,BSPublications,2003.
2	Signals and Systems-A.V. Oppenheim, A.S. Willsky and S.H. Nawab,PHI,2ndEdn,1997
Reference Books:	
1.	Principles of Linear Systems and Signals- B P Lathi, Oxford University Press
2.	Signals and Systems–TK Rawat, Oxford University press,2011
3	Signals&Systems-Simon Haykin and VanVeen,Wiley,2ndEdition,2007
e-Resources	
1.	https://archive.nptel.ac.in/courses/108/104/108104100/
2.	https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2102	PC	3	--	--	3	30	70	3 Hrs.
ELECTRONIC DEVICES AND CIRCUITS								
(For ECE)								
Course Objectives:								
1.	To learn and understand the basic concepts of semiconductor physics.							
2.	Study the physical phenomena such as conduction, transport mechanism and electrical characteristics of different diodes.							
3.	To learn and understand the application of diodes as rectifiers with their operation and characteristics with and without filters are discussed.							
4	Acquire knowledge about the principle of working and operation of Bipolar Junction Transistor and Field Effect Transistor and their characteristics.							
5	To learn and understand the purpose of transistor biasing and its significance.							
6	Small signal equivalent circuit analysis of BJT and FET transistor amplifiers andcompare different configurations							
Course Outcomes: Student will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the basic concepts of semiconductor physics and analyze the characteristics and operation of Semiconductor diode							K4
2.	Analyze the working principle, characteristics of special Semiconductor Diodes and Diode Circuits							K4
3.	Analyze the working principle, characteristics and biasing circuits of BJT							K4
4.	Analyze the performance of low frequency small signal BJT amplifier models.							K4
5.	Analyze the working principle, characteristics and small signal model of FET.							K4
SYLLABUS								
UNIT-I (12 hrs)	Review of Semiconductor Physics: Mobility and Conductivity, Intrinsic and extrinsic semiconductors, Fermi level in intrinsic and extrinsic Semiconductors. Junction Diode Characteristics: Open circuited p-n junction, Biased p-n junction, p-n junction diode, current components in p-n junction Diode, V-I Characteristics, temperature dependence on V-I characteristics, Diode resistance, Diode capacitance.							
UNIT-II (8 hrs)	Special Semiconductor Devices: Zener Diode, Breakdown mechanisms, Zener diode applications, Varactor Diode, Tunnel Diode, UJT, Construction, operation and V-I characteristics. Diode Circuits: Basic Rectifier setup, half wave rectifier, full wave rectifier, bridge							

	rectifier, derivations of characteristics of rectifiers, Filters, Capacitor filter, comparison of various rectifier circuits in terms of ripple factors.
UNIT-III (8 hrs)	Transistor Characteristics: Junction transistor, transistor current components, transistor equation in CB configuration, transistor as an amplifier, characteristics of transistor in Common Base, Common Emitter and Common Collector configurations, punch through/ reach through, Photo transistor, typical transistor junction voltage values. Transistor Biasing and Thermal Stabilization: Need for biasing, operating point, load line analysis, BJT biasing- methods, basic stability, fixed bias, collector to base bias, self-bias, Stabilization against variations in V_{BE}, I_C, and β, Stability factors, (S, S', S''), Derivation for S only, Bias compensation, Thermal runaway, Thermal stability.
UNIT-IV (8 hrs)	Small Signal Low Frequency Transistor Amplifier Models BJT: Two port network, Transistor hybrid model, determination of h-parameters, generalized analysis of transistor amplifier model using h-parameters, Analysis of Common Base, Common Emitter and Common Collector amplifiers using exact and approximate analysis, Comparison of transistor amplifiers.
UNIT-V (12 hrs)	Field Effect Transistor: Field Effect Transistor types, Junction Field Effect Transistor operation, characteristics, small signal model of Junction Field Effect Transistor, Transconductance, Drain Resistance, amplification factor and relation between them. Metal Oxide Semiconductor Field Effect Transistors, its V-I and transfer characteristics.
Textbooks:	
1.	Millman's Electronic Devices and Circuits- J. Millman, C. C. Halkias and Satyabrata Jit, Mc-Graw Hill Education, 4 th edition, 2015.
2.	Electronics devices & circuit theory- Robert L. Boylestad and Louis Nashelsky, Pearson, 11 th edition, 2015.
Reference Books:	
1.	Basic Electronics-Principles and Applications, Chinmoy Saha, Arindam Halder, Debarati Ganguly, Cambridge University Press.
2.	Electronic Devices and Circuits - David A. Bell, Oxford University Press, 5 th edition, 2008.
3.	Electronic Devices and Circuits- S. Salivahanan, N. Suresh Kumar, Mc-Graw Hill, 5 th edition, 2022.
4.	Millman's Integrated Electronics-J. Millman, C. Halkias, and Ch. D. Parikh, Mc-Graw Hill Education, 2 nd Edition, 2009.
5.	Fundamentals of Microelectronics-Behzad Razavi, Wiley, 3 rd edition, 2021.
e-Resources	
1.	https://nptel.ac.in/courses/117/103/117103063/
2.	https://www.satishkashyap.com/2013/03/video-lectures-on-electron-devices-by.html

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2103	PC	3	--	--	3	30	70	3 Hrs.

SWITCHING THEORY and LOGIC DESIGN

(For ECE)

Course Objectives:

1.	To solve a typical number base conversion and analyze new error coding techniques.
2.	Theorems and functions of Boolean algebra and behavior of logic gates
3.	To optimize logic gates for digital circuits using various techniques.
4.	Boolean function simplification using Karnaugh maps and Quine-McCluskey methods
5.	To understand concepts of combinational circuits.
6.	To develop advanced sequential circuits

Course Outcomes: Student will be able to

S.No	Outcome	Knowledge Level
1.	Summarize various number systems to operate different Codes.	K2
2.	Apply the concept of Boolean Algebra and minimization techniques to reduce Boolean functions.	K3
3.	Design various combinational logic circuits for specified statements.	K4
4.	Analyse various Sequential circuits using Flip Flops.	K3
5.	Design counters and state machines by applying the knowledge of synchronous and asynchronous sequential circuits.	K4

SYLLABUS

UNIT-I (10 hrs)	REVIEW OF NUMBER SYSTEMS & CODES: Representation of numbers of different radix, conversion from one radix to another radix, 1's and 2's complement of binary numbers, Gray code, 4 bit Codes: BCD, Excess-3, 2421, 8421 codes. Error detection & correction codes: parity checking, even parity, odd parity, Hamming code. BOOLEAN THEOREMS AND LOGIC OPERATIONS: Boolean theorems, principle of complementation & duality, De-morgan theorems, Logic operations: Basic logic operations -NOT, OR, AND, Universal Logic operations, EX-OR, EX- NOR operations, Standard SOP and POS Forms, NAND-NAND and NOR-NOR realizations, Realization of three level logic circuits.
UNIT-II (10 hrs)	MINIMIZATION TECHNIQUES: Minimization and realization of switching functions using Boolean theorems, K-Map (up to 4 variables) and tabular method (Quine-mccluskey method) with only four variables and

	single function. COMBINATIONAL LOGIC CIRCUITS DESIGN: Design of Half adder, full adder, half subtractor, full subtractor, applications of full adders, 4-bit adder-subtractor circuit, carry look-ahead adder circuit, Design code converts using Karnaugh method and draw the complete circuit diagrams.
UNIT-III (10 hrs)	COMBINATIONAL LOGIC CIRCUITS DESIGN USING MSI & LSI: Design of encoder, decoder, multiplexer and de-multiplexers, Implementation of higher order circuits using lower order circuits. Realization of Boolean functions using decoders and multiplexers. Design of Priority encoder, 4-bit digital comparator and seven segment decoder. INTRODUCTION OF PLD's: PLDs: PROM, PAL, PLA -Basics structures, realization of Boolean functions, Programming table.
UNIT-IV (10 hrs)	SEQUENTIAL CIRCUITS I: Classification of sequential circuits (synchronous and asynchronous), operation of NAND & NOR Latches and flip-flops; truth tables and excitation tables of RS flip-flop, JK flip-flop, Race around problem, JK Master-Slave flipflop, T flip-flop, D flip-flop with reset and clear terminals, Conversion from one flip-flop to another flip-flop, Design of registers - shift register, bi-directional shift register, universal shift register.
UNIT-V (10 hrs)	SEQUENTIAL CIRCUITS II : Design of ripple counters, design of synchronous counters, Johnson counter, ring counter. Analysis and design of synchronous circuits using state diagrams, state tables, reduction of state tables, Analysis of clocked sequential circuits Mealy to Moore conversion and vice-versa. Realization of sequence generator, Design of Clocked Sequential Circuit to detect the given sequence (with overlapping or without overlapping) up to three states.
Textbooks:	
1.	Switching and finite automata theory Zvi.Kohavi, Niraj.K.Jha 3 rd Edition, Cambridge University Press, 2009.
2.	Fundamentals of Logic Design by Charles H. Roth Jr, Jaico Publishers, 2006
Reference Books:	
1.	Digital electronics by R S Sedha. S. Chand & company limited, 2010
2.	Switching Theory and Logic Design by A. Anand Kumar, PHI Learning Pvt Ltd, 2016.
3.	Digital logic applications and design by John M Yarbough, Cengage Learning, 2006.
4.	TTL 74-Series data book.
e-Resources	
1	https://www.geeksforgeeks.org/minimization-of-boolean-functions/
2	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/#blg
3	https://www.cs.ou.edu/~fagg/classes/ame3623_s05/lectures/class_sequential.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E	Exam
B23EC2104	PC	--	--	3	1.5	30	70	3 Hrs.
ELECTRONIC DEVICES AND CIRCUITS LAB								
(For ECE)								
Course Objectives:								
1.	To analyze the modeling, characteristics and electrical parameters of Diode, BJT, JFET and MOSFET.							
2.	To illustrate the concepts of biasing in BJT, JFET and MOSFET. .							
3.	To illustrate the application of diode in rectifiers and regulated power supply.							
4.	To analyze single stage amplifier circuits using equivalent circuits.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the concepts of different electronic devices to verify their characteristics and measure the important parameters.							K3
2.	Analyze the performance of rectifier circuits with and without filters.							K4
3.	Analyze the performance of BJT and FET amplifier circuits.							K4
4.	Simulation and Design of small electronic circuits using BJT and FET.							K5
List of Experiments :(Minimum of Ten Experiments has to be performed)								
1.	Study and Analyze V-I Characteristics of Semiconductor Diode (Ge & Si), LED and Zener Diode.							
2.	Regulation characteristics of zener diode.							
3.	Rectifiers (without and with c-filter) Part A: Half-wave Rectifier Part B: Full-wave Rectifier							
4.	Transistor act as a switch							
5.	Transistor Biasing							
6.	Study and Analyze the Characteristics of BJT in CE Configuration and Determination of h-parameters.							
7.	CE Amplifier							
8.	CRO Operation and its Measurements.							
9.	Emitter Follower-CC Amplifier							

10.	Study and Analyze the Characteristics of BJT in CB Configuration and Determination of h-parameters.
11.	FET Characteristics(CS Configuration) Part A: Drain Characteristics Part B: Transfer Characteristics
12.	FET-CS Amplifier
Reference Books:	
1.	Lab manual
e-Resources:	
1.	http://vlabs.iitkgp.ac.in/be/#
2.	https://nptel.ac.in/courses/117/103/117103063/



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2105	PC	--	--	3	1.5	30	70	3 Hrs.
SWITCHING THEORY AND LOGIC DESIGN LAB								
(For ECE)								
Course Objectives:								
1	To outline the basics of Digital electronics, Boolean algebra, and be able to design the simple logic circuits and verify the functionality.							
2	Design combinational and sequential logic circuits using digital ICs.							
3	This laboratory course enables students to get practical experience in design, assembly and evaluation of switching theory and logic design. students use a digital trainer kit and Vivado emulator to test their electronic designs.							
Course Outcomes: On completion of this laboratory course, the students will be able to								
S.No	Outcome							Knowledge Level
1	Analyze and design basic combinational logic circuits using Digital IC' and HDL programming.							K4
2	Analyze basic sequential logic circuits using Digital IC's and HDL programming.							K4
LIST OF EXPERIMENTS								
1	Verify the basic logic gates and implement the following using basic logic gate ICs. a. Full adder. b. Full subtractor.							
2	Verify the logic functions of 8:1 Multiplexer using IC 74151 and 1:16 Demultiplexer using IC 74154. a. Implement the given function using 8:1 Multiplexer b. Convert 1:16 Demultiplexer into a 4:16 Decoder and implement the given function.							
3	Verify the given Code convertors using digital trainer kit a. Implement BCD to Seven segment decoder driver using IC 74147 and Seven segment Display b. 8 to 3 Priority encoder using IC 74148.							
4	Verify the operation of following flip-flops using Digital Trainer Kit a. SR-latch & RS-latch b. JK- flip flop using IC 7476 along with direct inputs. c. D- flip flop d. T- flip flop							

5	Design and verify the following counters using Digital Trainer Kit a. Mod-16 Ripple counter using IC7493 b. Mod-8 Synchronous counter using 7476 and 7408 ICs c. Decade (Mod-10) counter using IC7493
6	Verify the functioning of 5-bit shift register IC7496 using Digital Trainer Kit.
7	Introduction to HDL and Modelling Techniques
8	Develop the HDL Code for Logic gates using dataflow modelling and verify its functionality using Vivado Simulator
9	Develop the HDL Code for Full adder using dataflow modelling and verify its functionality using Vivado Simulator
10	Develop the HDL Code for 8:1 multiplexer using dataflow modelling and verify its functionality using Vivado Simulator.
11	Develop the HDL Code for 4-bit ALU using dataflow modelling and verify arithmetic and logical operations and verify its functionality using Vivado Simulator.
12	Develop the HDL Code for SR Latch, D Latch and T Latch using dataflow modelling and verify its functionality using Vivado Simulator.
13	Develop the HDL Code for 4-bit Ripple Carry Adder using dataflow modelling and verify its functionality using Vivado Simulator.
14	Develop the HDL Code for 4-Bit Shift Register using dataflow modelling and verify its functionality using Vivado Simulator.
Reference Books:	
1	Switching & Finite Automata theory – ZviKohavi and Neeraj K Jha, ,3rd Edition, Cambridge.
2	Digital Design – Morris Mano, PHI, 3rd Edition.
3	Digital Design With an Introduction to the Verilog HDL By <u>M. Morris Mano</u> , <u>Michael D. Ciletti</u>
4	Verilog HDL Synthesis: A Practical Primer by Bhasker, Jayaram
5	Digital Systems Design Using Verilog-Charles H. Roth.
6	Digital Systems Design Using VHDL-Charles H. Roth
7	Digital ICs Lab Manual
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105185/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2106	SEC	--	1	2	2	30	70	3 Hrs.
DATA STRUCTURES USING PYTHON								
(For ECE)								
Course Objectives:								
1.	To introduce the basics of Python Programming Language							
2.	To learn about basic data science libraries.							
3.	To learn about search and sorting operations							
4.	To be familiar with stacks, queues and trees							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Use the data structures like lists, sets, dictionaries, and tuples to store collections of related data							K3
2.	Utilize python libraries to performs mathematical operations and draw charts							K3
3.	Perform search and sort of numbers using algorithms							K3
4.	Implement basic Stack and Queue operations							K3
LIST OF EXPERIMENTS								
Part-A: Basic Python Programs								
1	Basic programs: a. Python program to add two numbers b. Maximum of two numbers in Python c. Python program to print all Prime numbers in an Interval d. Python program to check whether a number is Prime or not e. Program to print ASCII Value of a character							
2	Array Programs: a. Python Program to find sum of array b. Python Program to find largest element in an array c. Python Program for array rotation							
3	List Programs: a. Python program to print even numbers in a list b. Python program to find N largest elements from a list c. Python program to Sort the values of first list using second list							
4	Dictionary Programs:							

	a. Ways to sort list of dictionaries by values in Python – Using itemgetter b. Python – Sort Dictionary key and values List
5	Tuple Programs: a. Python – Maximum and Minimum K elements in Tuple b. Sort a list of tuples by second Item
6	Numpy, Scipy, matplotlib, pandas: a. Basic mathematical operations- Numpy b. Single Integrals-scipy c. Matplotlib d. Pandas
Part-B: Data Structures using python	
1	Write a program to implement basic Queue operations
2	Write a program to implement basic stack operations
3	Write a program to find a number using linear and binary search
4	Write a program to implement a. Bubble sort b. Selection sort
5	Write a program to implement Binary tree
Text Books:	
1.	Automate the Boring Stuff with Python by Al Sweigart
2.	Problem Solving with Algorithms and Data Structures using Python" by Brad Miller and David Ranum
3.	Python for everybody by Dr. Charles R. Severance
4.	Python Data Science Handbook: Essential tools for working with data by Jake VanderPlas, Oreilly books
5.	Learning Python by Mark Lutz, Oreilly books
Reference Books:	
1	“Learning Python” by Mark Lutz and David Ascher 2nd Edition.
2	“Think Python: How to Think Like a Computer Scientist”, by Allen B. Downey 2nd Edition.
3	"Data Structures and Algorithms in Python" by Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser
E-Resources	
1.	https://www.coursera.org/learn/python
2.	Joy of computing using Python : https://nptel.ac.in/courses/106/106/106106182/
3.	Python for data science: https://nptel.ac.in/courses/106/106/106106212/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2102	MC	2	--	--	--	30	--	--
ENVIRONMENTAL SCIENCE								
(Common to CIC, CSG, CSE, ECE and EEE)								
Course Objectives: The objective of the course is to impart								
1.	Overall view on natural resources.							
2.	Awareness on ecosystem and its services.							
3.	Various environmental challenges induced due to unplanned anthropogenic activities.							
4.	Consciousness on the social issues, environmental legislation and global treaties							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1	Describe natural resources and their interaction							K2
2	Illustrate ecosystem types, biodiversity and conservation strategies							K2
3	Summarize contaminants of environment and preventive methods							K2
4	Explain protection of environment by employing constitutional provisions							K2
5	Explain global scenario of surroundings and social conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary Nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources							
UNIT-II (8 Hrs)	Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)							

	Biodiversity and Its Conservation : Introduction and Definition - genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.
Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.
6	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.

Reference Books:	
1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k





SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. II - Semester							
ELECTRONICS & COMMUNICATION ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2201	Managerial Economics and Financial Analysis	HS	2	0	0	2	30	70	100
B23EC2201	Linear Control Systems	ES	3	0	0	3	30	70	100
B23EC2202	Electromagnetic Waves and Transmission Lines	PC	3	0	0	3	30	70	100
B23EC2203	Electronic Circuit Analysis	PC	3	0	0	3	30	70	100
B23EC2204	Analog Communications	PC	3	0	0	3	30	70	100
B23EC2205	Signals and Systems Lab	PC	0	0	3	1.5	30	70	100
B23EC2206	Electronic Circuit Analysis lab	PC	0	0	3	1.5	30	70	100
B23EC2207	Hardware Description Language Laboratory	SEC	0	1	2	2	30	70	100
B23EC2208	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2201	English Proficiency	MC	2	0	0	-	30	-	30
TOTAL			17	1	10	21	300	630	930

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2201	HS	2	--	--	2	30	70	3Hrs.
MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS								
(Common to AIDS, CSE, CIC, CSG, CSIT, CE, ECE, EEE, ME)								
Course Objectives: Students are expected to								
1.	Understand the concept and nature of Managerial Economics, its relationship with other disciplines, the Concept of Demand and Demand forecasting							
2.	Familiarize about the Concepts of Cost and Break-Even Analysis							
3.	Learn about accounting cycle and preparation of Financial Statements							
4.	Understand the nature of markets and the Pricing Methods							
5.	Know the concept of Capital, Sources of Raising Finance and Depreciation							
Course Outcomes: At the end of the course the student will be able to								
S.No	Outcome							Knowledge Level
1.	Interpreting the importance of Managerial Economics, demand analysis and methods of demand forecasting							K2
2.	Describe about the usefulness of Cost Analysis and Break Even Analysis							K2
3.	Apply the principles of accounting to convert the transactions and events into Journal, Ledger and Trail balance							K3
4.	Compute the results of Business by preparing Final Accounts							K3
5.	Illustrate the nature of markets and pricing theories							K2
6.	Explain the Types of capital, their sources and importance & estimation of Depreciation							K2
SYLLABUS								
UNIT-I (12 Hrs)	Introduction to Managerial Economics and demand Analysis: Managerial Economics: Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics. Demand Analysis: Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of Demand Forecasting and its Methods.							
UNIT-II (8 Hrs)	Cost Analysis: Importance of cost analysis, Types of Cost- Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; Elements of costs – Material, Labour, Expenses; Methods of costing - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. Break-even analysis: Determination of Break-							

	even Point Applications, Assumptions and Limitations of Break-even analysis (Theory only).
UNIT-III (12 Hrs)	Introduction to Financial Accounting: Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (Final Accounts with Simple adjustments).
UNIT-IV (8 Hrs)	Introduction to Markets & Pricing Policies Market Structures: Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. Pricing: Importance of pricing and its meaning; Methods of Pricing: Cost Based -Full cost, Mark-up, Marginal & Break-even; Demand Based -Penetrating, Skimming; Competition Based -Going rate, Sealed Bid, Discount; Internet Pricing -Flat-rate, Usage sensitive.
UNIT-V (8 Hrs)	Capital & Depreciation: Types of Capital-Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, Medium term and Long term. Depreciation – Meaning, Importance and causes of depreciation; Methods of Depreciation-Straight line and Diminishing Balancing methods (Theory only).
Text Books:	
1.	AR Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N. Appa Rao, Dr. P. Vijaya kumar: Managerial Economics and Financial Analysis, Cengage Publications, NewDelhi
Reference Books:	
1.	Dr.B.Kuberudu & T.V.Ramana :Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K.Gupta & R.K.Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2201	ES	3	--	--	3	30	70	3 Hrs.
LINEAR CONTROL SYSTEMS								
(For ECE)								
Course Objectives: Students will learn								
1.	The modelling of linear systems using transfer functions and obtain transfer functions using block diagrams and signal flow graphs.							
2.	The significance of time response and find it for system analysis in transient and steady state.							
3.	The concept of stability and know different techniques of stability analysis.							
4.	The concept of frequency domain analysis, Bode plots, Polar plots							
5.	The concept of state space modeling and analysis.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Model electrical and mechanical physical systems by applying laws of physics and derive transfer functions from block diagrams & Signal Flow Graphs.							K3
2.	Analyze systems in time domain for transient and steady-state behavior.							K4
3.	Analyze the stability of a system by RH criterion and Root locus.							K4
4.	Analyze the behavior of system using frequency response plots.							K4
5.	Model and analyze the LTI system using state space approach.							K4
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION TO CONTROL SYSTEMS Open loop and closed loop systems, Transfer Function models of linear Systems-Modelling of Electrical & mechanical Systems, Block Diagram representation of Control Systems – Block Diagram Reduction, Signal Flow Graph Representation of Control Systems, Mason’s gain formula, Feedback Characteristics of Control Systems.							
UNIT-II (10 Hrs)	TIME DOMAIN ANALYSIS OF CONTROL SYSTEMS Time Response of First and Second Order Systems with Standard Input Signals, Time Domain Specifications of Second Order Systems, Steady State Error, Steady State Error Constants-Basic Control Actions- Effects of Integral and Derivative Control actions.							
UNIT-III (10 Hrs)	STABILITY ANALYSIS OF CONTROL SYSTEMS Concept of Stability, Routh-Hurwitz Criterion, Relative Stability Analysis, The Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus.							

UNIT-IV (10 Hrs)	FREQUENCY DOMAIN ANALYSIS OF CONTROL SYSTEMS Frequency Response -Bode Plots- Log Magnitude versus Phase Plots, Polar Plots – Frequency Domain specifications -Correlation between Time and Frequency Responses, Stability in Frequency Domain- Nyquist Stability Criterion - Assessment of Relative Stability, Gain Margin and Phase Margin.
UNIT-V (10 Hrs)	STATE SPACE ANALYSIS Concept of state, State Variables and State Models, State space models for LTI electrical Systems, Phase variable form and diagonal canonical form, Conversion between Transfer Function models and State space Models, Solution to the State Equation, State Transition Matrix, Concept of Controllability and Observability.
Text Books:	
1.	I. J. Nagrath and M. Gopal, “Control Systems Engineering”, New Age International Publishers (6 th Edition).
2.	Norman S.Nise, ‘Control systems Engineering ‘, Wiley publications (7 th Edition)
Reference Books:	
1.	Katsuhiko Ogata, “Modern Control Engineering” PHI (4 th Edition).
2.	Richard C. Dorf and Robert H. Bishop, „Modern Control Systems“, Addison-Wesley Publishers (8 th Edition)



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2202	PC	3	--	--	3	30	70	3 Hrs.
ELECTROMAGNETIC WAVES AND TRANSMISSION LINES								
(For ECE)								
Course Objectives:								
1.	To impart knowledge about the characteristics and computation of static electric and magnetic fields and their implications.							
2.	To introduce Maxwell’s equations and their applications.							
3.	To illustrate the concept of Electromagnetic Wave Propagation and transmission line theory.							
Course Outcomes: At the end of the course the students will be able to								
S.No	Outcome							Knowledge Level
1.	Compute transmission line parameters for different transmission line configurations.							K3
2.	Determine electric field intensity using coulomb’s law and Gauss law for different charge distributions.							K3
3.	Determine magnetic field intensity using Biot-Savarts Law and Ampere’s Circuital Law for different current distributions.							K3
4.	Apply Maxwell’s equations to describe the behaviour of time varying electromagnetic fields.							K3
5.	Apply Maxwell’s equations to describe the EM wave propagation in dielectric and conducting media.							K3
SYLLABUS								
UNIT-I (8 hrs)	Transmission Lines - I : Types, Parameters, T & π Equivalent Circuits, Transmission Line Equations, Primary & Secondary Constants, Expressions for Characteristic Impedance, Propagation Constant, Phase and Group Velocities, Infinite Line, Lossless lines, distortion less lines.							
UNIT-II (10 hrs)	Transmission Lines – II: Input Impedance Relations, Reflection Coefficient, VSWR, Average Power, Shorted Lines, Open Circuited Lines, and Matched Lines, Low loss radio frequency and UHF Transmission lines, UHF Lines as Circuit Elements, Smith Chart – Construction and Applications, Quarter wave transformer, Single Stub Matching.							
UNIT-III (15 hrs)	Electrostatics: Coulomb’s Law, Electric Field Intensity, Electric Flux Density, Gauss Law and Applications, Electric Potential, Maxwell’s Two Equations for Electrostatic Fields,							

	Energy Density, Convection and Conduction Currents, Dielectric Constant, Poisson's and Laplace's Equations; Capacitance – Parallel Plate, Coaxial Capacitors.
UNIT-IV (15 hrs)	Magnetostatics: Biot-Savart Law, Ampere's Circuital Law and Applications, Magnetic Flux Density, Maxwell's Two Equations for Magnetostatic Fields, Magnetic Scalar and Vector Potentials, Forces due to Magnetic Fields, Ampere's Force Law, Inductances and Magnetic Energy. Maxwell's Equations (Time Varying Fields): Faraday's Law and Transformer EMF, Inconsistency of Ampere's Law and Displacement Current Density, Maxwell's Equations in Different Final Forms and Word Statements, Boundary Conditions at the interface between two media.
UNIT-V (10 hrs)	EM Wave Characteristics: Wave Equations for Conducting and Perfect Dielectric Media, Uniform Plane Waves – Definition, All Relations Between E & H, Sinusoidal Variations, Wave Propagation in Lossy dielectrics, lossless dielectrics, free space, wave propagation in good conductors, skin depth, wave propagation in good dielectrics, Polarization & Types, Poynting Vector and Poynting Theorem.
Textbooks:	
1.	Principles of Electromagnetics - N.O.Sadiku, Oxford University Press, 4th edition.
2.	EM Waves and Radiating Systems – E.C.Jordan and K.G.Balmain, Printice Hall India
Reference Books:	
1.	Engineering Electromagnetics – William H. Hayt, John A. Buck, Jaleel M. Akhtar, TMH, 9th edition, 2020.
2.	Electromagnetic field theory and Transmission lines – G.S.N.Raju, Pearson Education Pvt.Ltd.
3.	Electromagnetic Field Theory and Transmission Lines - Gottapu Sasibhushana Rao - Wiley
e-Resources	
1.	https://nptel.ac.in/courses/108104087
2.	https://www.youtube.com/watch?v=xrxO1FhUaWg&list=PLs5_Rtf2P2r5AJvcEuBEwdur_dUVojW-

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2203	PC	3	--	--	3	30	70	3 Hrs.
ELECTRONIC CIRCUIT ANALYSIS								
(For ECE)								
Course Objectives: The main objectives of this course are:								
1.	Learn and understand the purpose of cascading of single stage amplifiers and derive the overall voltage gain.							
2.	Analyze the effect of negative feedback on amplifier characteristics and derive the characteristics.							
3.	Learn and understand the basic principle of oscillator circuits and perform the analysis of different oscillator circuits.							
4	Compare and analyze different Power amplifiers like Class A, Class B, Class C, Class AB and other types of amplifiers.							
5	Analyze different types of tuned amplifier circuits.							
Course Outcomes: At the end of the course the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze small signal high frequency transistor amplifier using BJT and FET.							K4
2.	Design multistage amplifiers using BJT and FET and Differential amplifier using BJT.							K4
3.	Analyze negative Feedback amplifiers							K4
4.	Derive the expressions for frequency of oscillation and condition for oscillation of RC and LC oscillators and their amplitude and frequency stability concept.							K3
5.	Analyze the performance of power amplifiers and tuned amplifiers							K3
SYLLABUS								
UNIT-I (10 Hrs)	Small Signal High Frequency Transistor Amplifier models: BJT: Transistor at high frequencies, Hybrid- π common emitter transistor model, Hybrid π conductance, Hybrid π capacitances, validity of hybrid π model, determination of high-frequency parameters in terms of low-frequency parameters, CE short circuit current gain, current gain with resistive load, cut-off frequencies, frequency response and gain bandwidth product. FET: Analysis of common Source and common drain Amplifier circuits at high frequencies.							
UNIT-II	Multistage Amplifiers: Classification of amplifiers, methods of coupling,							

(10 Hrs)	cascaded transistor amplifier and its analysis, analysis of two stage RC coupled amplifier, high input resistance transistor amplifier circuits and their analysis- Darlington pair amplifier, CasCourse Code amplifier, Boot-strap emitter follower, Differential amplifier using BJT.
UNIT-III (10 Hrs)	Feedback Amplifiers: Feedback principle and concept, types of feedback, classification of amplifiers, feedback topologies, Characteristics of negative feedback amplifiers, Generalized analysis of feedback amplifiers, Performance comparison of feedback amplifiers, Method of analysis of feedback amplifiers.
UNIT-IV (8 Hrs)	Oscillators: Oscillator principle, condition for oscillations, types of oscillators, RC-phase shift and Wien bridge oscillators with BJT and FET and their analysis, Generalized analysis of LC Oscillators, Hartley and Colpitt's oscillators using BJT, Frequency and amplitude stability of oscillators.
UNIT-V (10 Hrs)	Power Amplifiers: Classification of amplifiers(A to H), Class A power Amplifiers, Class B Push-pull amplifiers, Complementary symmetry push pull amplifier, Class AB power amplifier, Class-C power amplifier, Thermal stability and Heat sinks. Tuned Amplifiers: Introduction, Q-Factor, small signal tuned amplifier, capacitance single tuned amplifier, double tuned amplifiers, , staggered tuned amplifiers.
Textbooks:	
1.	Integrated Electronics- J.Millman and C.C.Halkias, Tata McGraw-Hill, 1972.
2.	Electronic Devices and Circuits Theory –Robert L.Boylestad and Louis Nashelsky, Pearson/PrenticeHall, Tenth Edition, 2009.
Reference Books:	
1.	Electronic Circuit Analysis and Design –Donald A.Neaman, McGrawHill, 2010.
2.	Microelectronic Circuits-Sedra A.S. and K.C. Smith, Oxford University Press, Sixth Edition, 2011.
3.	Electronic Circuit Analysis-B.V.Rao, K.R.Rajeswari, P.C.R.Pantulu, K.B.R.Murthy, Pearson Publications
4.	Electronic Devices and Integrated Circuits – B.P. Singh, Rekha, Pearson publications, 2006
e-Resources	
1	https://www.uotechnology.edu.iq/dep-eee/lectures/3rd/Electronic/Analog%2520electronic/part2.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2204	PC	3	--	--	3	30	70	3 Hrs.
ANALOG COMMUNICATIONS								
(For ECE)								
Course Objectives:								
1.	Familiarize with the fundamental concepts of communication systems and various techniques of Analog modulation and demodulation of signals.							
2.	To provide a good understanding of the behavior of analog communications in the presence of noise.							
3.	To classify and discuss different types of transmitters and receivers as applicable to analog communication systems.							
4.	Familiarize with basic techniques of generating and demodulating Pulse modulated signals.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze the concepts of Amplitude Modulation and Demodulation techniques.							K4
2.	Analyze different types of Amplitude Modulation and Demodulation techniques.							K4
3.	Analyze the concepts of generation and detection of Angle Modulated signals.							K4
4.	Illustrate the various functional blocks of Radio transmitters and Radio receivers with different sections.							K2
5.	Compute the noise performance of Analog Modulation techniques and explain the concepts of Pulse Analog Modulation and Demodulation techniques							K3
SYLLABUS								
UNIT-I (8 hours)	Amplitude Modulation: Introduction to Fourier transform, Introduction to communication system, Need for modulation, Frequency Division Multiplexing, Amplitude Modulation, Time domain and Frequency domain descriptions, Single tone modulation, Power relations in AM waves, Generation of AM waves: Square law Modulator, Switching modulator, Detection of AM Waves: Square law detector, Envelope detector, Related problems.							
UNIT-II (8 hours)	DSB & SSB Modulation: Double sideband suppressed carrier modulator: Time domain and frequency domain description, Generation of DSBSC Waves: Balanced Modulator, Ring Modulator, Detection of DSBSC Waves: Coherent detection, Quadrature Null							

	Effect, COSTAS Loop, Squaring Loop. Single sideband suppressed carrier modulator: Time domain and Frequency domain description, Generation of SSBSC Waves: Frequency discrimination method, Phase discrimination method, Demodulation of SSB Waves: Coherent Detection. Vestigial sideband modulation: Time domain description, Frequency domain description, Generation of VSB Modulated wave, Envelope detection of a VSB Wave pulse Carrier, Comparison of different AM Techniques, Applications of different AM Systems, Related problems.
UNIT-III (8 hours)	Angle Modulation: Introduction, Basic concept of phase modulation, Frequency Modulation: Single tone frequency modulation, Spectrum Analysis of Sinusoidal FM Wave, Narrow band FM, Wide band FM, Constant Average Power, Transmission bandwidth of FM Wave, Generation of FM Waves: Direct Method, Indirect Method, Detection of FM Waves: Balanced Frequency discriminator, Zero crossing detector, Phase locked loop, Comparison of FM & AM, Related problems.
UNIT-IV (8 hours)	Radio Transmitters: Classification of Transmitters, AM Transmitter, Effect of feedback on performance of AM Transmitter, FM Transmitter: Variable reactance type and Phase modulated FM Transmitter, Frequency stability in FM Transmitter. Radio Receivers: Receiver Types: Tuned radio frequency receiver, Super heterodyne receiver, RF section and Characteristics, Frequency changing and tracking, Intermediate frequency, AGC, FM Receiver, Amplitude limiting, Comparison of FM & AM Receivers, Communication Receivers, Extension of super heterodyne principle and additional circuits.
UNIT-V (8 hours)	Noise: Review of noise and noise sources, Noise figure, Noise in Analog communication Systems: Noise in DSB & SSB Systems, Noise in AM System and Noise in Angle Modulation Systems, Threshold effect in Angle Modulation System, Pre-emphasis & De-emphasis. Pulse Analog Modulation: Types of Pulse modulation, PAM (Single polarity, double polarity), PWM: Generation & Detection of PWM, PPM: Generation and Detection of PPM, Time Division Multiplexing, TDM Vs FDM.
Textbooks:	
1.	Principles of Communication Systems, H Taub, D L Schilling, Gautam Sahe, TMH, 4th Edition, 2017.
2.	Communication Systems, Simon Haykin, Michael Moher, Wiley, 5th Edition, 2009.
3.	Modern Digital and Analog Communication Systems, B.P.Lathi, Zhi Ding, Hari Mohan Gupta, Oxford University Press, 4th Edition, 2017
Reference Books:	
1.	Electronics & Communication Systems, George Kennedy, Bernard Davis, S R M Prasanna, TMH, 6th Edition, 2017.
2	Communication Systems, R P Singh, S D Sapre, TMH, 3rd Edition, 2017.

3	Communication Systems (Analog and Digital), Dr. Sanjay Sharma, Katson Books, 7th Reprint Edition, 2018.
e-Resources	
1	http://nptel.ac.in/courses/117102059
2	https://nptel.ac.in/courses/117/105/117105143/
3	https://www.scribd.com/document/266137872/sanjay-sharma-pdf .
4	http://bayanbox.ir/view/914409083519889086/Book-Modern-Digital-And-AnalogCommunication-Systems-4th-edition-by-Lathi.pdf .
5.	https://soaneemrana.org/onewebmedia/ELECTRONICS%20COMMUNICATION%20SYSTEM%20BY%20GEORGE%20KENNEDY.pdf



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2205	PC	--	--	3	1.5	30	70	3hrs
SIGNALS AND SYSTEMS LAB								
(For ECE)								
Course Objectives:								
1.	The primary goal of this lab course is to provide a thorough Understanding and Analysis of Signals and Systems in Time / Frequency domains using Computational, Programming and Visualization features of MATLAB software. Students will develop a skill with MATLAB programming to solve signals and linear systems problems in engineering.							
Course Outcomes: On the successful completion of this laboratory course the students will be able to								
S.No	Outcome							Knowledge level
1.	Compute basic signal operations on CT/DT signals in time-domain.							K3
2.	Analyze the frequency domain characteristics of CT/DT signals/systems using Fourier tools.							K4
3.	Illustrate the sampling theorem, the dynamic system simulation using Laplace and z- transforms.							K3
LIST OF PROGRAMS								
1	MATLAB Software & Toolboxes a) Introduction: Environment, Variables/Operators, Basic Commands, Matrix Algebra, Scripts, Functions, Graphics b) Signal Processing Toolbox Functions and GUIs							
2	Generation of Basic Continuous-Time and Discrete-Time Signals a) CT / DT Signals: DC, Sinusoidal, Exponential, Complex exponential, Unit impulse, Unit step & Sinc functions b) Natural Signals: Speech and Image c) Random Signals & Gaussian Noise.							
3	Transformations & Operations on CT/ DT Signals a) Amplitude scaling, DC biasing, Time shifting, Time reversal, Time scaling, b) Addition & Multiplication, Even/Odd decomposition.							
4	Convolution & Correlation of Signals a) CT / DT Convolution and Applications of Convolution b) Correlation & Applications of Correlation.							
5	Frequency Domain Representation of Periodic Signals: Fourier Series a) Frequency analysis of CT periodic signals.							
6	Frequency Domain Representation of CT Aperiodic Signals: Fourier Transform							

	a) Fourier Transform (FT) and Properties
7	Frequency Domain Representation of DT Aperiodic Signals: DTFT a) Discrete-Time Fourier Transform (DTFT)
8	Frequency Domain Representation of LTI Systems a) Frequency Response of CT-LTI Systems b) Frequency Response of DT-LTI Systems
9	CT / DT LTI System Simulation a) Response of CT/DT LTI systems to different inputs: Sinusoidal, Unit impulse & Unit step signals
10	Laplace Transform, Z-Transform & Applications a) Laplace Transform (LT), Z-Transform (ZT) and Inverse Transforms b) Convolution & Solving the LCC Differential and Difference equations.
11	Sampling Theorem, Signal Reconstruction & Aliasing a) Sampling Theorem Verification & Aliasing Effect.
12	SIMULINK & BLOCKSETS a) Introduction to the Block diagram (Graphical) based Dynamic System Modeling and Simulation, Blocksets, Simulink Library Browser. b) Build Models to Demonstrate: Generation and Operations on signals including Convolution, Fourier transform, Filtering and Solving the Differential and Difference equations.
References:	
1.	SS Lab Manual
2	Textbook: Signals and Systems Using MATLAB by Luis F. Chaparro
3.	Textbook: Digital Signal Processing Using MATLAB, 3rd ed. by Vinay K. Ingle John G. Proakis
4.	https://www.mathworks.com/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2206	PC	--	--	3	1.5	30	70	3hrs
ELECTRONIC CIRCUIT ANALYSIS LAB								
(For ECE)								
Course Objectives:								
1	To empower students to understand the design and working of BJT amplifiers, Oscillators, Power amplifiers and tuned amplifiers							
2	Model the electronic circuits using Multisim Software.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Simulation and Design of different voltage and power amplifiers to verify their characteristics and measure the important parameters.							K4
2	Simulation and Design of different oscillator circuits.							K4
LIST OF EXPERIMENTS								
HARDWARE								
1	Frequency Response of RC Coupled Amplifier.							
2	Frequency Response of Negative Feedback Amplifier.							
3	Colpitts Oscillator.							
4	RC Phase Shift Oscillator.							
5	Wein Bridge Oscillator.							
6	Hartley Oscillator.							
7	Complementary Symmetry class-AB power amplifier							
8	Tuned Voltage Amplifier.							
SOFTWARE								
1	Frequency Response of RC Coupled Amplifier							
2	Frequency Response of Negative Feedback Amplifier							
3	Colpitts Oscillator							
4	RC Phase Shift Oscillator							
5	Wein Bridge Oscillator							
6	Hartley Oscillator							
7	Complementary Symmetry class-AB power amplifier							
8	Tuned Voltage Amplifier.							
Reference Books:								
1.	ECAD Laboratory Manual							
e-Resources:								
1.	https://www.orcad.com/resources/orcad-tutorials							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2207	SEC	--	1	2	2	30	70	3hrs
HARDWARE DESCRIPTION LANGUAGE LABORATORY								
(For ECE)								
Course Objectives:								
1	Familiarize students to proficiently develop HDL programs for digital design applications.							
2	Enhance students' comprehension of simulation and synthesis methodologies employed in digital design.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1	Develop HDL (Hardware Description Language) programs for various digital design applications, incorporating combinational circuits, sequential circuits, and finite state machines effectively.							K3
2	Comprehend simulation and synthesis methodologies used in digital design, enabling them to optimize digital circuits for performance, resource utilization, and timing constraints.							K4
LIST OF EXPERIMENTS								
Develop HDL(VHDL/Verilog HDL/System Verilog) Code for the following and Verify functionality using Xilinx Vivado/Intel Altera. Perform simulation tests to ensure correct behavior under various input conditions/Test benches, and verify synthesis results. (Any 12 experiments)								
1	Introduction to HDL and Modelling Techniques							
2	Logic Gates using data flow, behavioral and gate level modelling.							
3	Adders & Subtractors using data flow, behavioral and structural modelling.							
4	4-bit Binary Adder and Subtractor using structural modelling.							
5	Multiplexers using dataflow,behavioral and structural modelling.							
6	Basic logic gates (AND, OR, NOT, XOR, etc.) using 2:1 multiplexers in behavioral style of modelling.							
7	Full adder using 2:1 Multiplexer and 4:1 Multiplexer in behavioral style of modelling.							
8	Latches/Flip Flops using behavioral style of modelling							
9	Decoders using NAND/NOR gates in behavioral style of modelling.							
10	Encoders & Priority Encoders using behavioral style of modelling.							
11	4-bit BCD synchronous counter using behavioral style of modelling.							
12	8-bit ALU using behavioral style of modelling.							
13	Finite State Machine (FSM) that detects a random sequence in an input sequence of bits.							
14	8-bit Random Access Memory(RAM) /Read-Only Memory (ROM) using behavioral style of modelling.							
15	Assignment-1.							
16	Assignment-2.							

Reference Books:	
1	Switching & Finite Automata theory – ZviKohavi and Neeraj K Jha, ,3rd Edition, Cambridge.
2	Verilog HDL Synthesis : A Practical Primer by Bhasker, Jayaram
3	Digital Systems Design Using Verilog-Charles H. Roth.
4	<u>Digital Design with Verilog - Course (nptel.ac.in)</u>
5	Digital Systems Design Using VHDL-Charles H. Roth.
6	VHDL: Programming By Example By Perry, Douglas L
7	Hardware Verification with System Verilog: Mike Mintz, Robert Ekendahl
8	Lab Manual.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EC2208	ES	1	--	2	2	30	70	3 Hrs.
DESIGN THINKING & INNOVATION								
(Common to all Programmes of Engineering)								
Course Objectives:								
1.	Bring awareness on innovative design and new product development.							
2.	Explain the basics of design thinking.							
3.	Familiarize the role of reverse engineering in product development.							
4.	Train how to identify the needs of society and convert into demand.							
5.	Introduce product planning and product development process.							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Define the concepts related to design thinking.							K1
2.	Explain the fundamentals of Design Thinking and innovation.							K2
3.	Apply the design thinking techniques for solving problems in various sectors.							K3
4.	Analyse to work in a multidisciplinary environment.							K4
5.	Evaluate the value of creativity.							K5
SYLLABUS								
UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.							
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.							
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.							
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.							

	Activity: Importance of modeling, how to set specifications, Explaining their own product design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes. Activity: How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2201	MC	2	--	--	--	30	--	--
ENGLISH PROFICIENCY								
(Common to CSE, CE, ECE, EEE, CIC, CSD).								
Course Outcomes: After completion of this course, students will be able to								
Sl. No	Outcome							Knowledge Level
1.	Interpret IELTS & TOEFL listening comprehension texts.							K2
2.	Demonstrate essential speaking skills in academic, professional, and real-life contexts.							K2
3.	Interpret the written discourse by applying effective reading strategies.							K2
4.	Construct coherent and cohesive paragraphs, e-mails, letters, and essays.							K3
SYLLABUS								
UNIT-I	Listening Skills Listening as a receptive skill Listening skills for IELTS, TOEFL, Duolingo, PTE, and other competitive examinations.							
UNIT-II	Speaking Skills JAM/ Extempore Debate / Group Discussion Presentation Skills							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive Reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive essays for GRE, TOEFL & IELTS) E-Mail / Letter Writing Resume Writing							
UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking							
Text Books:								

1.	Interchange (5 th edition) by Jack C.Richards, CUP, 2017.
Reference Books:	
1.	Fundamentals of Technical Communication (1 st edition) by Meenakshi Raman, Sangeeta Sharma of OUP, 2014.
2.	The Oxford Guide to Writing and Speaking (3 rd edition) by John Seely OUP, 2013.
3.	Effective Technical Communication (2 nd edition) by M. Ashraf Rizwi. TataMcGrawhill, 2017.
e-Resources:	
1	BBC Learning English - Learn English with BBC Learning English - Homepage
2	Grammar Learn English (britishcouncil.org)
3	Duolingo English Test
4	IELTS Test Preparation Materials - Videos, Practice tests, Articles and More (idp.com)

